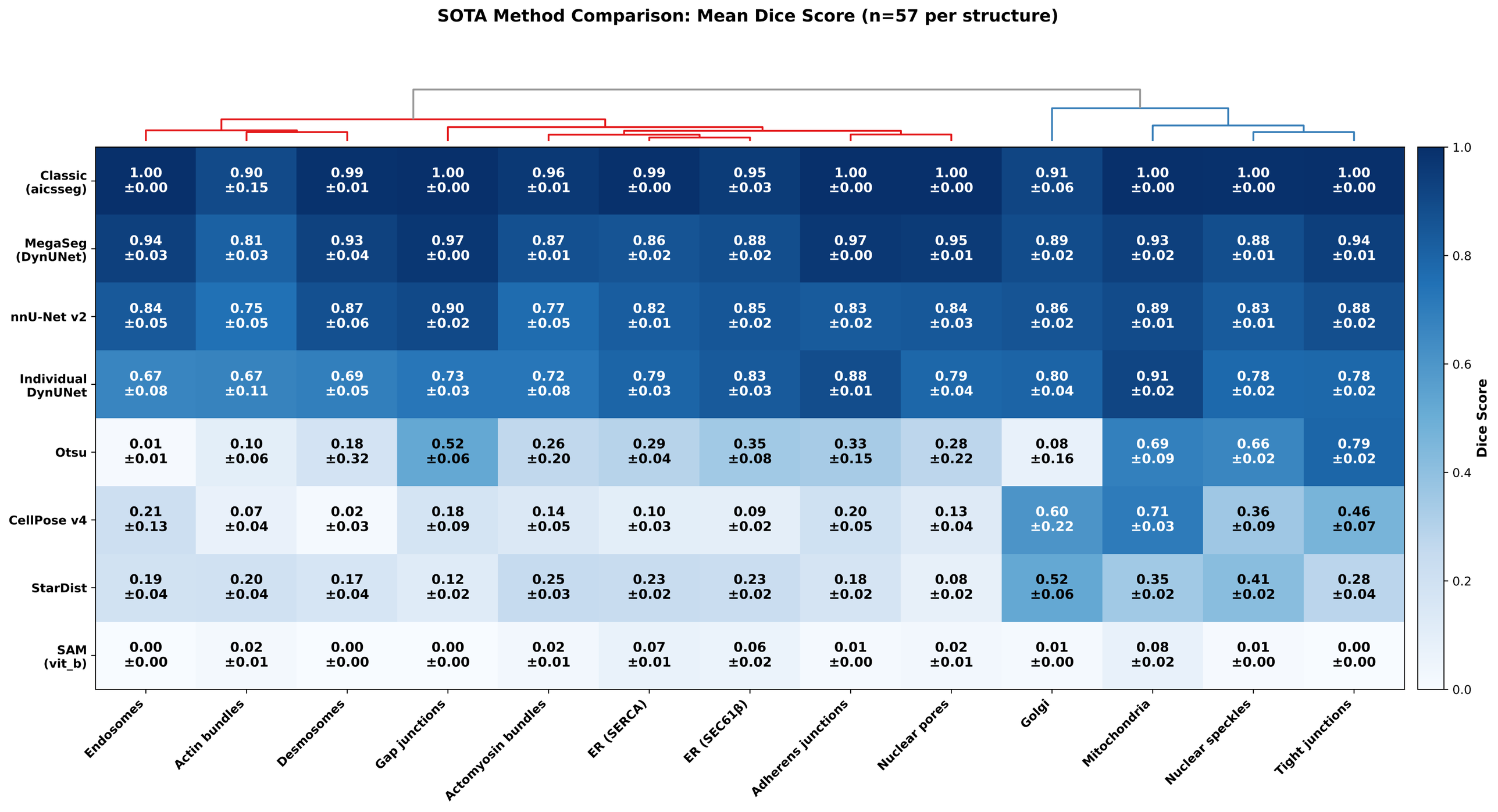

SOTA Method Comparison: Mean Dice Score (n=57 per structure)
Classic (aicsseg)
MegaSeg (DynUNet)
nnU-Net v2
Individual DynUNet
Otsu
CellPose v4
StarDist
SAM (vit_b)
Endosomes
Actin bundles
Desmosomes
Gap junctions
Actomyosin bundles
ER (SERCA)
ER (SEC61β)
Adherens junctions
Nuclear pores
Golgi
Mitochondria
Nuclear speckles
Tight junctions
Dice Score
1.0
0.8
0.6
0.4
0.2
0.0
1.00 ±0.00
0.90 ±0.15
0.99 ±0.01
1.00 ±0.00
0.96 ±0.01
0.99 ±0.00
0.95 ±0.03
1.00 ±0.00
1.00 ±0.00
0.91 ±0.06
1.00 ±0.00
1.00 ±0.00
1.00 ±0.00
0.94 ±0.03
0.81 ±0.03
0.93 ±0.04
0.97 ±0.00
0.87 ±0.01
0.86 ±0.02
0.88 ±0.02
0.97 ±0.00
0.95 ±0.01
0.89 ±0.02
0.93 ±0.02
0.88 ±0.01
0.94 ±0.01
0.84 ±0.05
0.75 ±0.05
0.87 ±0.06
0.90 ±0.02
0.77 ±0.05
0.82 ±0.01
0.85 ±0.02
0.83 ±0.02
0.84 ±0.03
0.86 ±0.02
0.89 ±0.01
0.83 ±0.01
0.88 ±0.02
0.67 ±0.08
0.67 ±0.11
0.69 ±0.05
0.73 ±0.03
0.72 ±0.08
0.79 ±0.03
0.83 ±0.03
0.88 ±0.01
0.79 ±0.04
0.80 ±0.04
0.91 ±0.02
0.78 ±0.02
0.78 ±0.02
0.01 ±0.01
0.10 ±0.06
0.18 ±0.32
0.52 ±0.06
0.26 ±0.20
0.29 ±0.04
0.35 ±0.08
0.33 ±0.15
0.28 ±0.22
0.08 ±0.16
0.69 ±0.09
0.66 ±0.02
0.79 ±0.02
0.21 ±0.13
0.07 ±0.04
0.02 ±0.03
0.18 ±0.09
0.14 ±0.05
0.10 ±0.03
0.09 ±0.02
0.20 ±0.05
0.13 ±0.04
0.60 ±0.22
0.71 ±0.03
0.36 ±0.09
0.46 ±0.07
0.19 ±0.04
0.20 ±0.04
0.17 ±0.04
0.12 ±0.02
0.25 ±0.03
0.23 ±0.02
0.23 ±0.02
0.18 ±0.02
0.08 ±0.02
0.52 ±0.06
0.35 ±0.02
0.41 ±0.02
0.28 ±0.04
0.00 ±0.00
0.02 ±0.01
0.00 ±0.00
0.00 ±0.00
0.02 ±0.01
0.07 ±0.01
0.06 ±0.02
0.01 ±0.00
0.02 ±0.01
0.01 ±0.00
0.08 ±0.02
0.01 ±0.00
0.00 ±0.00